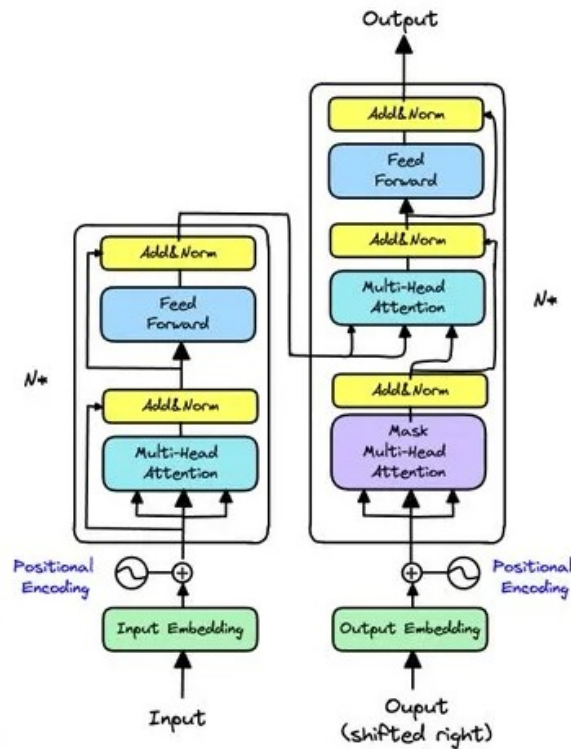


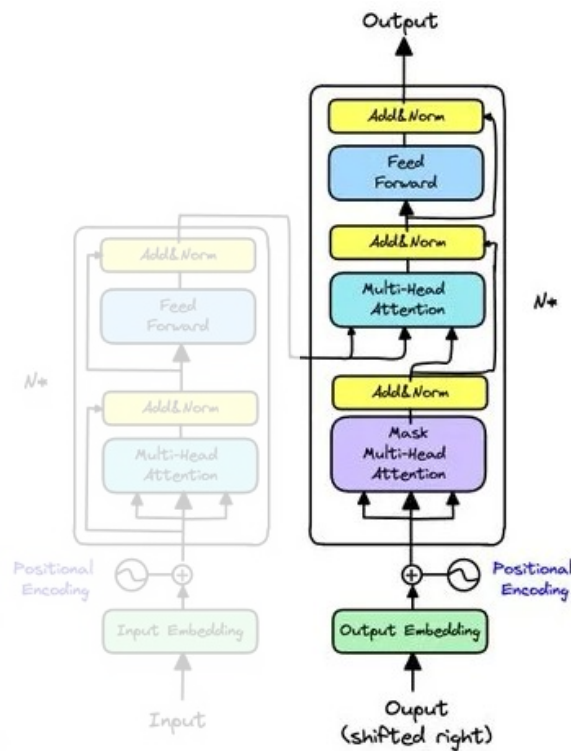
Transformers

Transformers- Origin



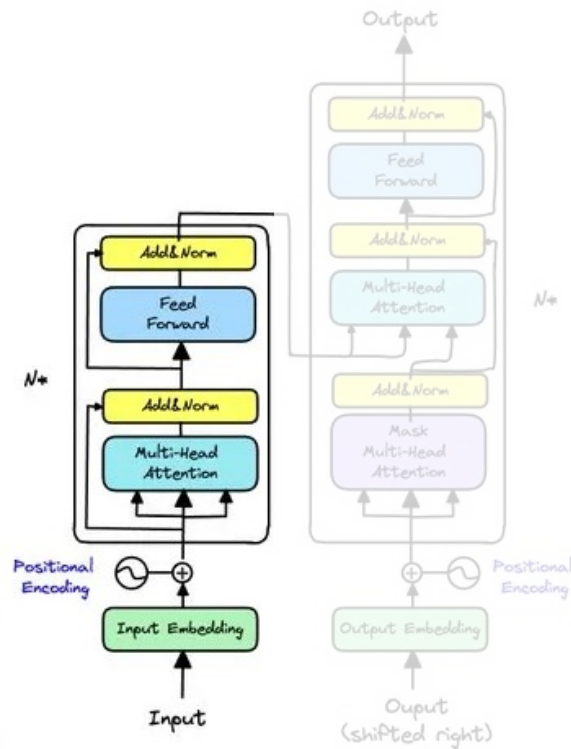
- This is the original transformer
- It was designed to be a sequence to sequence model
- It translated from 1 language to another

Transformers-Decoder



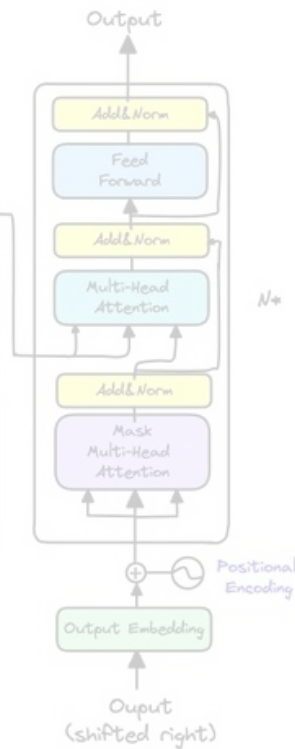
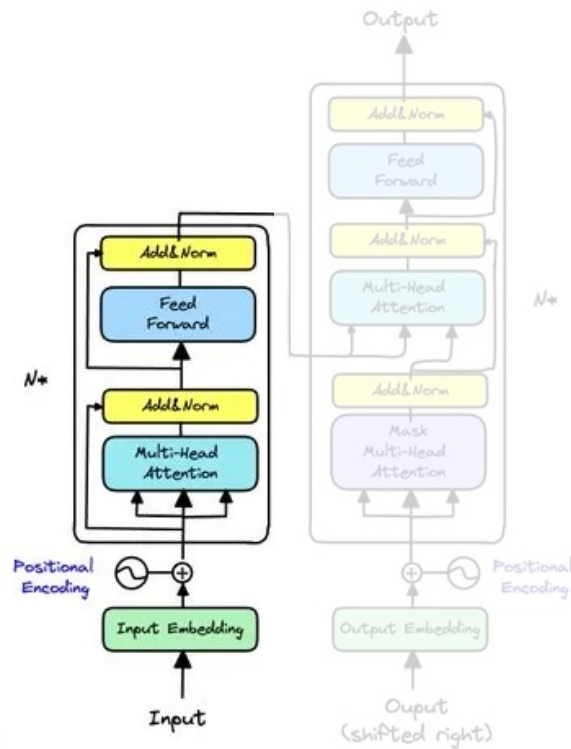
- The right side is the **decoder**
- Predicts the next token in a sequence.
- It works by taking a sequence of token IDs, processing them to understand the context, and then generates output **token by token (serial process)**
- The core function of generative LLMs like ChatGPT.
- We will not do much with these

Transformers-Encoder



- The left side is the **encoder**
- Creates a contextualized representation of the input (**parallel process**)
- Used to generate context aware embeddings (RAG)
The river **bank**
The money in the **bank**
- Also can be fine tuned for sentiment analysis, question answering, and text classification
- Uses Self Attention

Transformers-Encoder



- Examples (BERT, ModernBERT, SBERT...there are a lot)
- We will mostly use encoders for the remainder of this course